

- Proof that the length of the Key must be as long as the message M .

By contradiction,

let $\{0,1\}^{m_1}$ be the set of values that the key K can take

let $\{0,1\}^{m_2}$ be the set of values that the message M can take

assume that $m_1 < m_2$, and for any $m \in \{0,1\}^{m_2}$,
 $IP[M=m] \neq 0$

Step 1. Suppose that Oscar observes a ciphertext c
 Oscar tries all the possible keys to decrypt c

From c , Oscar gets

$$\begin{cases} m_1 \\ m_2 \\ \vdots \\ m_{2^{m_1}} \end{cases}$$

Step 2. Since $2^{m_1} < 2^{m_2}$, there exists a message m such that $IP[M=m | C=c] = 0$

Step 3 $\underbrace{IP[M=m | C=c]}_{=0} \neq \underbrace{IP[M=m]}_{\neq 0}$

This means that M and C are not independent : contradiction

Summary (one-time pad)

- The key must only be used once
- The key must be as long as the message
- You need to exchange a secret key
- Provide "unconditional" security, i.e., it provides security against computationally unbounded adversaries

Remark: this contrasts with all the other security protocols that we are going to see next. This contrasts with complexity-based cryptography, which assumes that adversaries are computationally limited.

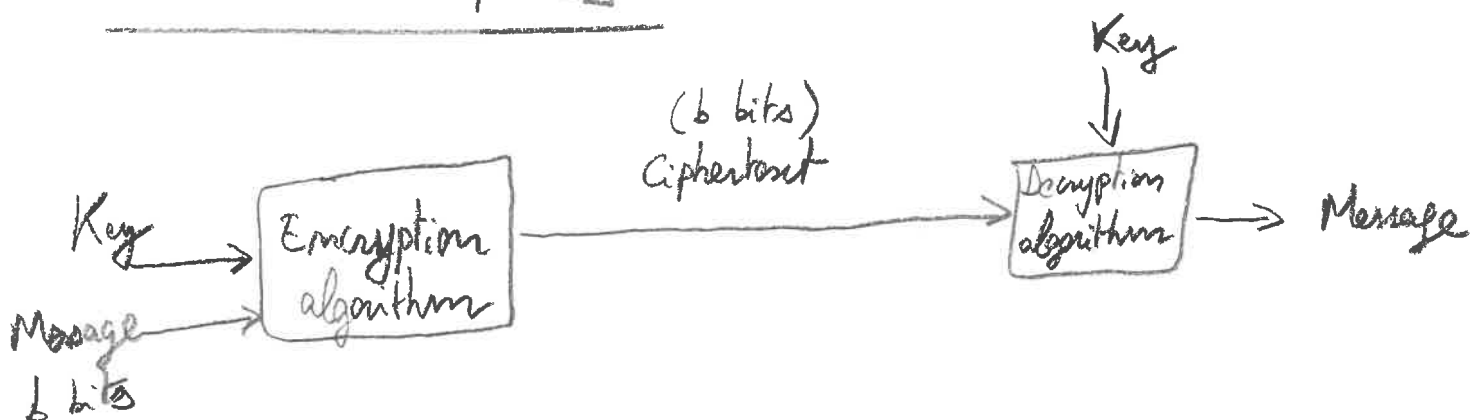
Remark: Caesar cipher $\rightarrow$ substitution cipher $\rightarrow$ One-time pad

Design a solution

$\hookrightarrow$ analyze the performance $\rightarrow$ refine your solution



III. Block ciphers



For the Caesar cipher, Substitution cipher, and one-time pad, $b=1$

Remark The encryption algorithm should be invertible.
How many invertible mappings exist between $\{0,1\}^b$ and $\{0,1\}^b$?
 $2^b!$
This would require a long key (the key would index the mapping)

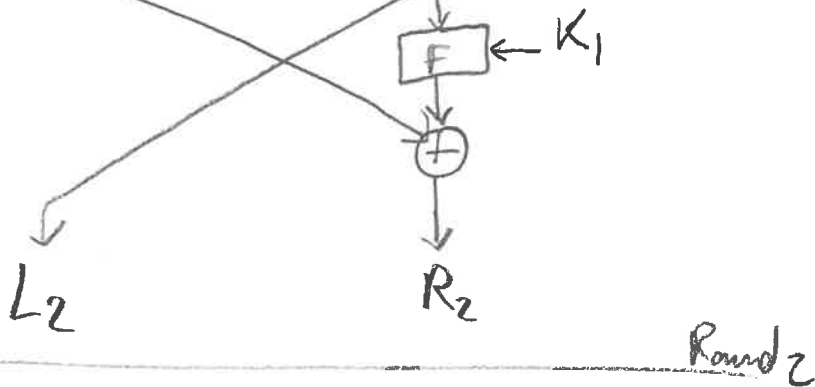
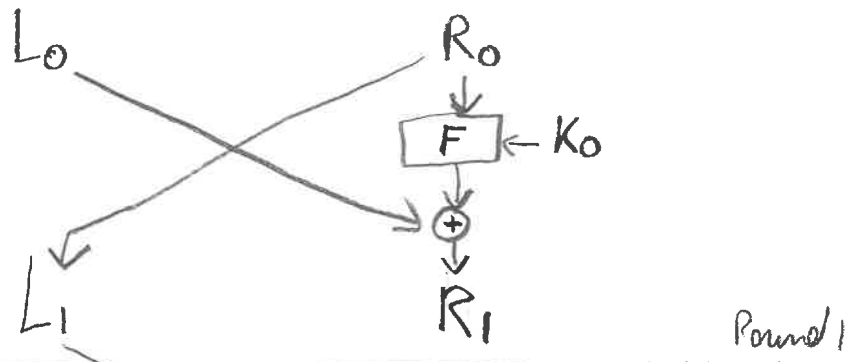
III. A Feistel cipher

Idea: Approximate an ideal block cipher (which requires a key alphabet size equal to $2^b!$) by a combination of simple ciphers that require a small key length k .

The Feistel cipher relies on substitutions and permutations:

Encryption protocol of Feistel cipher : Two sequences of bits of equal size

Plaintext is a sequence of bits written (L_0, R_0)



$\vdots$

$\vdots$

L_m R_m **Round m**

short description

For $i = 0 \dots m-1$

$$L_{i+1} = R_i$$

$$R_{i+1} = F(R_i, K_i) \oplus L_i$$

(L_m, R_m) is the ciphertext

Decryption protocol of Feistel cipher :

For $i = m-1, \dots, 0$

$$R_i = L_{i+1}$$

$$L_i = R_{i+1} \oplus F(R_i, K_i) = R_{i+1} \oplus F(L_{i+1}, K_i)$$

Parameters and design features

- Block size : Security generally improves with larger block size
larger block sizes increase the encryption/decryption speed.
- Key size : _____
- Number of rounds : _____
- Ease of analysis : not easy to analyze

III. B. DES (Data encryption Standard)

DES uses the Feistel cipher with $m = 16$ rounds

Adopted by NIST in 1977

Broken in 1999 $\rightarrow$ NIST issued new directives
Triple DES
(Three consecutive applications of DES)

DES 56 bit keys

Time required with 10^{13} decryption/s
1 hour

3DES 168 bit keys

$5.8 \cdot 10^{29}$ years